OpenAI’s GPT-3

Impact on the Future of Legal AI

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Who is OpenAI?

OpenAI is an AI research and deployment company based in San Francisco, California.

Founders

Elon Musk, Sam Altman, Ilya Sutskever, Greg Brockman, Wojciech Zaremba, John Schulman

Mission

To ensure artificial general intelligence* benefits all of humanity.

***Artificial general intelligence** (**AGI**) is the hypothetical intelligence of a machine that has the capacity to understand or learn any intellectual task that a human being can. (Wikipedia)

OpenAI’s GPT-3

OpenAI released a natural language processing (NLP) application programming interface (API) in June 2020.

- The underlying model (aka GPT-3) is not publicly accessible
- An user can access the model <u>only</u> through the website or API
- The underlying model cannot be modified by users

GPT-3 is widely considered the current state-of-the-art language model.

Why is GPT-3 State-of-the-Art?

- General Purpose
- Training Data 100% Unlabeled
- Trained on a Huge Amount of Data
- Very Large Model Size (175B model parameters)
- Requires Very Small Amount of User Data to Adapt to a New Task

General Purpose Model

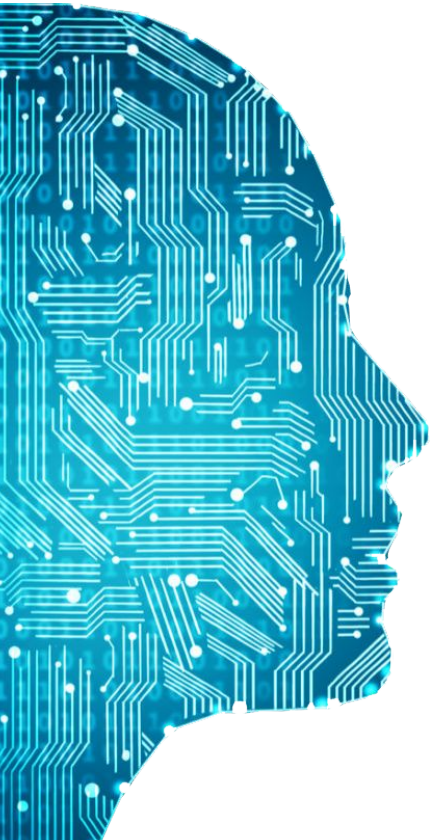
An AI system generally can do only one task:

- Siri/Echo: speech recognition
- Netflix: movie recommendation
- Gmail Auto-Complete: predictive text completion

GPT-3 is a general-purpose model, “allowing users to try it on virtually any English language task.”

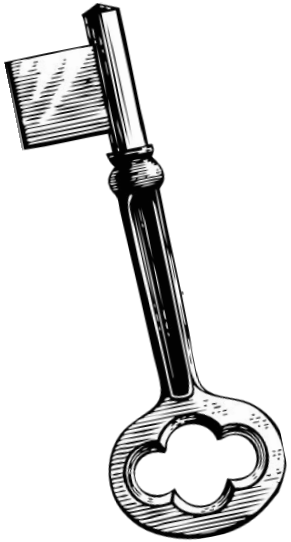
https://openai.com/blog/openai-api/

Training Data 100% Unlabeled



- Most AI models rely on a large amount of labeled data (from thousands to billions)
  - Labeling often needs to be done by humans.
    - Time-consuming & costly
- GPT-3, by contrast, was trained on unlabeled data
  - Data across the web contains “implicit” labels, not requiring human relabeling for a particular task.

Large-Scale General Purpose Model: Key to Success for AI in Legal?



- Slow AI development in Legal is due to lack of expert-labelled legal datasets
- GPT-3 was trained on datasets that included legal contracts available on the web
  - Probably millions or more
- Legal AI may leverage GPT-3 to:
  - Minimize human labeling
  - Increase accuracy

Example Use-Cases in Legal

- Text Completion
- Q&A
- Summarization

Use-Case #1: Text Completion

- Alex inputs two paragraphs from his college thesis about the history of corporations.
- GPT-3 automatically generates a third paragraph providing supporting evidence and citing “Santa Clara County v. Southern Pacific railroad Company (1886)”, the focus of Alex’s thesis.

Playground

Parse unstructured data

Input:

150 years ago, corporations were small, limited in number, heavily regulated, and severely restricted in their activities. Today, the presence of large multinational corporations is the defining feature of modern economic life. In the past 150 years, a revolution has occurred that has seen corporations go from small organizations chartered to fulfill specific functions, to massive institutions that again, have come to define modern economic life. What allowed corporations to grow and to become our most common form of business enterprise were a series of changes in the law throughout the 19th and 20th centuries in both state and federal law.Beginning in the 1850's, states, which had hitherto granted corporate charters under only specific conditions and which placed a slew of regulations on any chartered corporation, began to pass much more liberal corporate statutes. Encouraged by rapidly growing railroad companies, the desire to attract corporate investment, and outright bribery, state governments became increasingly permissive in their regulation of corporations throughout the late 19th century. In what Horowitz has called a “race to the bottom,” state governments such as New Jersey ad Delaware eventually passed corporate statues with virtually none of the regulation or limitations that had characterized corporate law only 50 years before. As corporations moved their headquarters to states with favorable legal climates, they were able to grow in size, power, and function, and morph into the kinds of large multi-faceted corporate enterprises we see today.

Output (generated by AI):

The other major legal change that occurred during the 19th century was the change in the status of corporations from entities that existed solely to fulfill a public function (and thus, not a natural person) to natural persons. In the early 19th century, courts ruled that corporations could not sue or be sued, and that shareholders did not have any liability for the corporation's debts. But as the 19th century progressed, courts began to reverse these holdings, and to rule that corporations were in fact natural persons and should be treated as such. The Supreme Court, in its decision in the 1886 case Santa Clara County v. Southern Pacific Railroad, went so far as to state that corporations were “persons” for purposes of the Fourteenth Amendment, which was enacted after the Civil War to protect the rights of the recently freed slaves.Corporations have also changed how they are structured, with most now being incorporated as either a C or an S corporation. C corporations, which make up the majority of corporations, are still chartered by the states, and are still regulated by the states, but are no longer subject to double taxation (i.e. taxation on the corporation's profits, and then again on the shareholders when the profits are distributed as dividends). C corporations are also subject to strict corporate governance requirements, and to the fiduciary duties of directors and officers.S corporations, on the other hand, are chartered by the states, but are not subject to the strict governance requirements or the double taxation of C corporations. While this lack of regulation is attractive to small businesses, it also makes S corporations particularly vulnerable to abuse. One of the major reasons that S corporations are often formed is to shield income from taxes. Thus, the use of S corporations to evade taxes has been called “the biggest tax scam in America.”While it is not always possible to trace the exact history of corporate law, there are three important Supreme Court cases that, along with state corporate statutes, are the foundation of modern corporate law. In Dodge v. Woolsey, decided in 1856, the Supreme Court ruled that corporations were separate legal entities from their shareholders, and that shareholders were not liable for corporate debts. In 1886, the Supreme Court ruled in Santa Clara County v. Southern Pacific Railroad that corporations were “persons” for purposes of the Fourteenth Amendment. In 1886, the Court ruled in Union Pacific v. United States that corporations were “persons” for purposes of the Contract Clause of the Constitution.

Use-Case #2: Q&A

- Spencer inputs the question “What is the Governing Law?” followed by several clauses from a contract.
- GPT-3 correctly replies with “The Governing Law is the law of the Commonwealth of Virginia.”

Playground

Q&A

Q: What is the Governing Law?
Company shall indemnify, defend and hold Reseller and its officers, directors, employees, shareholders,agents, successors and assigns, harmless from and against any claim, demand lawsuit, cause of action or losses of any nature whatsoever, including reasonable attorneys' fees, suffered or incurred by Reseller or any of them, arising out of, (i) any claim that the Company Products infringe any U.S. patent or copyright or that the Company Products incorporate any misappropriated trade secrets or (ii) any claim for injury or other loss to any person arising from the use of Company Products. Additionally,Reseller shall indemnify, defend and hold Company and its officers, directors, employees, shareholders, agents, successors and assigns, harmless from and against any claim, demand lawsuit, cause of action or losses of any nature whatsoever, including reasonable attorneys' fees, suffered or incurred by Company or any of them, arising out of, (i) any claim that the Reseller's Activities infringe any U.S. Business law or that the Reseller's Activities violate any U.S. government process, procedure or requirement or (ii) any claim for injury or other loss to any person arising from the use of Reseller's Activities.
If any provision of this Agreement is held to be illegal or unenforceable, that provision shall be limited or eliminated to the minimum extent necessary so that this Agreement shall otherwise remain in full force and effect and enforceable. Except as otherwise expressly provided herein, any provision of this Agreement may be amended or waived only with the written consent of both parties. This Agreement may be executed in separate counterparts, each of which so executed and delivered (including by emailed PDF) shall constitute an original, but all such counterparts shall together constitute one and the same instrument. This Agreement shall be governed by and construed under the laws of the Commonwealth of Virginia without regard to the conflicts of law provisions thereof. This Agreement supersedes all prior agreements or discussions between the parties with respect to the subject matter hereof.
Each party acknowledges that, in the course of performing its duties under this Agreement, it may obtain from the other party, certain business, technical or financial information, all of which is confidential and proprietary ("Proprietary Information"). The parties shall, at all times, both during the term of this Agreement and after its termination, keep in trust and confidence all such Proprietary Information of the other party, and shall not use such Proprietary Information other than in the course of its duties as expressly provided in this Agreement; nor shall a party or its employees disclose any such Proprietary Information of the other to any person without such other party's prior written consent. Neither party shall be bound by this Section with respect to information it can document has entered or later enters the public domain as a result of no act or omission of the receiving party, or is lawfully received by such party from third parties without restriction and without breach of any duty of nondisclosure by any such third party.

A: The Governing Law is the law of the Commonwealth of Virginia.

Use-Case #3: Summarization

- Anya inputs the following excerpt from a news article* about the California Consumer Privacy Act, with the question “What does the CCPA do?”
- GPT-3 responds with a reasonable two-sentence summary.

Playground

Summarize for a 2nd grader

What does the CCPA do?
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In early 2018 a California real estate developer spearheaded an effort to include a new privacy law — the Consumer Right to Privacy Act of 2018 — on the November 2018 California ballot. By June 2018, supporters of the initiative had gathered enough signatures to earn a place on the November ballot. In response, California legislators, working with representatives of affected California businesses and other interest groups, quickly negotiated and passed a substitute bill — The California Consumer Privacy Act of 2018 - the CCPA — in exchange for an agreement to drop the more restrictive text in the Consumer Right to Privacy Act from the November ballot. The California legislator passed what is considered to be the absolute toughest data privacy law in the United States. The California Consumer Privacy Act of 2018 was approved by the California State Governor on June 28, 2018, and goes into effect on January 1, 2020. The law applies to any business that has more than $25 million in revenue, or buys or sells the personal information of 50,000 or more consumers, or derives 50 percent or more of its annual revenue from selling consumers' personal information, and that does any amount of business in the State of California. The tone of the legislation is quite aggressive. The legislation specifically cites the March 2018 disclosure of the misuse of personal data by Cambridge Analytica. The legislation also references recent congressional hearings that followed which highlighted the fact that any personal information shared on the internet can be subject to considerable misuse and theft. This prompted the California legislature to move rapidly to protect Californians' right to privacy by giving consumers much more control of their personal information. The CCPA generally offers California consumers new statutory rights to learn what personal information Covered Businesses have collected, sold and disclosed, opportunities to opt-out of the sale of their personal information, and (uniquely) protection from "discrimination" in the form of reduced service or functionality for exercising those rights. For Covered Businesses, the CCPA specifies disclosure obligations and compliance procedures that are designed to help California consumers exercise these new statutory rights. A snapshot of the key provisions includes: Consumer Right to Know: Under the CCPA, similar to the new European regulations, California consumers have a right to request what personal information has been collected about them, as well as what personal information has been sold or otherwise disclosed about them. The CCPA requires Covered Businesses to comply with "verifiable requests" from consumers about the collection, sale, and disclosure of their personal information and outlines specific procedures and timelines that Covered Businesses must follow. Consumer Right to Delete: Under the CCPA, similar to the new the European regulations, California consumers have a right to request that their personal information be deleted. Covered Businesses must honor "verifiable" requests to delete consumer personal information, subject to several notable exceptions, including that a Covered Business need not delete personal information if maintaining the information is required to complete a transaction or provide a good or service. Consumer Opt-Out from Sale of Personal Information: Under the CCPA, California consumers are afforded the right to "opt-out" of the "sale" (which is broadly defined) of their personal information. Covered Businesses must provide notice of this right to consumers (including by providing a clear and conspicuous hyperlink entitled "Do Not Sell My Personal Information" on their websites) and must implement designated methods for consumers to opt-out (including a toll-free number and website address for opting-out). Covered Businesses must honor consumer opt-outs, and must wait 12 months before seeking re-authorization to sell their personal information. Consumer Opt-In for the Sale of Personal Information of Minors: Under the CCPA, the personal information of minors under the age of 13 may only be sold if the consumer's parent or guardian has affirmatively authorized (opted-in to) the sale. For minors aged 13-16, affirmative authorization is also required, but the consumer may provide the authorization. Non-Discrimination for Exercise of Consumer Rights: Under the CCPA, Covered Businesses are prohibited from discriminating against consumers based on their having exercised rights (i.e., opting out of collection or monetization of data) pursuant to the CCPA. A Covered Business cannot refuse to sell goods or provide services, charge different prices for such goods or services, or provide lower quality goods and services because a consumer exercises his or her rights under the CCPA. However, this requirement does not prohibit a Covered Business from charging different prices or providing different quality goods or services if the difference is "reasonably related" to the value of the personal information at issue. The CCPA is enforceable both by the Attorney General for the State of California and by private litigants. However, the Act contains technical terms regarding when and how a consumer can bring a private action under the statute. Notably, the terms provide Covered Businesses opportunities to cure certain instances of non-compliance. Key terms include: Enforcement by Attorney General: Violations of the CCPA are enforceable by the California Attorney General, which is authorized to pursue civil penalties of up to US$7,500 per violation. Limited Private Right of Action for Unauthorized Disclosure of Data: Consumers may bring a private right of action against Covered Businesses in connection with "certain unauthorized access and exfiltration, theft, or disclosure of a consumer's nonencrypted or nonredacted personal information" if the Covered Business has failed to implement and maintain reasonable security measures to protect such information. However, prior to commencing an action for statutory damages (US$100-$750 per incident), the consumer must provide the Covered Business with 30 days to cure the alleged violation and to respond with a written statement that the violation has been cured.
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Under the CCPA, similar to the new European regulations, California consumers have a right to request what personal information has been collected about them, as well as what personal information has been sold or otherwise disclosed about them. The CCPA requires Covered Businesses to comply with "verifiable requests" from consumers about the collection, sale, and disclosure of their personal information and outlines specific procedures and timelines that Covered Businesses must follow.

*Source: https://www.americanbar.org/groups/business_law/publications/committee_newsletters/bcl/2019/201902/fa_9/

Read More

Despite its success with the foregoing limited use cases, we are mindful of the mistakes that GPT-3 makes. Future Bytesized AI episodes will explore the results of using OpenAI's Beta API on the Atticus Dataset.

To learn more about OpenAI and GPT-3, visit their website, or these blog posts about the API and the unique strengths of the model.